Solution: Problem Challenge 3: Cycle in a Circular Array

Problem Statement

We are given an array containing positive and negative numbers. Suppose the array contains a number 'M' at a particular index. Now, if 'M' is positive we will move forward 'M' indices and if 'M' is negative move backwards 'M' indices. You should assume that the **array is circular** which means two things:

1. If, while moving forward, we reach the end of the array, we will jump to the first element to continue the movement.
2. If, while moving backward, we reach the beginning of the array, we will jump to the last element to continue the movement.

Write a method to determine **if the array has a cycle**. The cycle should have more than one element and should follow one direction which means the cycle should not contain both forward and backward movements.

Example 1:

```
Input: [1, 2, -1, 2, 2]
Output: true
Explanation: The array has a cycle among indices: 0 -> 1 -> 3 -> 0
```

Example 2:

```
Input: [2, 2, -1, 2]
Output: true
Explanation: The array has a cycle among indices: 1 -> 3 -> 1
```

Example 3:

```
Input: [2, 1, -1, -2]
Output: false
Explanation: The array does not have any cycle.
```

Solution

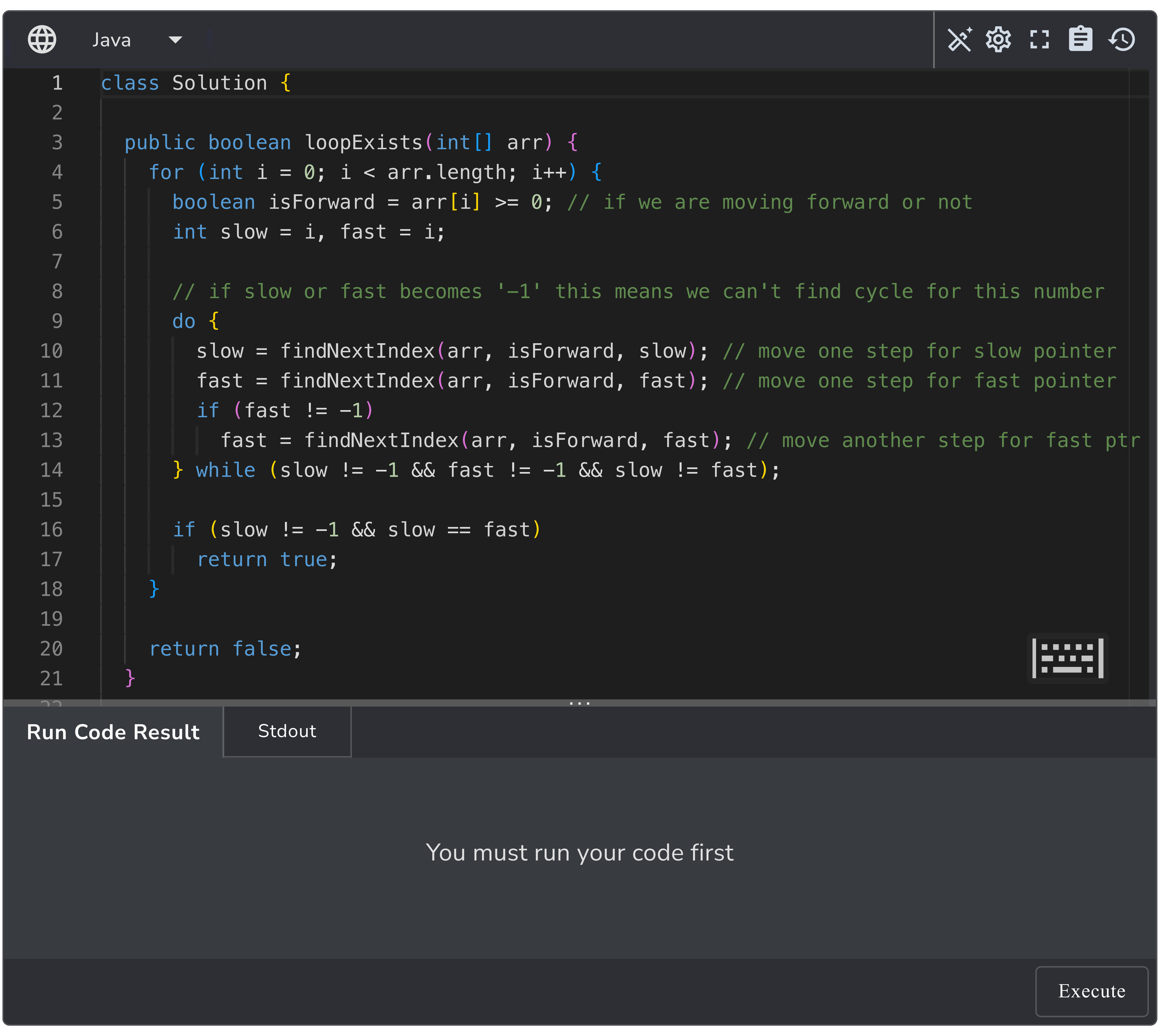
This problem involves finding a cycle in the array and, as we know, the **Fast & Slow pointer** method is an efficient way to do that. We can start from each index of the array to find the cycle. If a number does not have a cycle we will move forward to the next element. There are a couple of additional things we need to take care of:

1. As mentioned in the problem, the cycle should have more than one element. This means that when we move a pointer forward, if the pointer points to the same element after the move, we have a one-element cycle. Therefore, we can finish our cycle search for the current element.
2. The other requirement mentioned in the problem is that the cycle should not contain both forward and backward movements. We will handle this by remembering the direction of each element while searching for the cycle. If the number is positive, the direction will be forward and if the number is negative, the direction will be backward. So whenever we move a pointer forward, if there is a change in the direction, we will finish our cycle search right there for the current element.

Here is the visual representation of this algorithm for Example-1:

Code

Here is what our algorithm will look like:



```java
class Solution {

  public boolean loopExists(int[] arr) {
    for (int i = 0; i < arr.length; i++) {
      boolean isForward = arr[i] >= 0; // if we are moving forward or not
      int slow = i, fast = i;

      // if slow or fast becomes '-1' this means we can't find cycle for this number
      do {
        slow = findNextIndex(arr, isForward, slow); // move one step for slow pointer
        fast = findNextIndex(arr, isForward, fast); // move one step for fast pointer
        if (fast != -1)
          fast = findNextIndex(arr, isForward, fast); // move another step for fast ptr
      } while (slow != -1 && fast != -1 && slow != fast);

      if (slow != -1 && slow == fast)
        return true;
    }

    return false;
  }
```

You must run your code first

Time Complexity

The above algorithm will have a time complexity of $O(N^2)$ where 'N' is the number of elements in the array. This complexity is due to the fact that we are iterating all elements of the array and trying to find a cycle for each element.

Space Complexity

The algorithm runs in constant space O(1).

An Alternate Approach

In our algorithm, we don't keep a record of all the numbers that have been evaluated for cycles. We know that all such numbers will not produce a cycle for any other instance as well. If we can remember all the numbers that have been visited, our algorithm will improve to O(N) as, then, each number will be evaluated for cycles only once. We can keep track of this by creating a separate array, however, in this case, the space complexity of our algorithm will increase to O(N).

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